SAP AMS

Transition Management System

Complete Project Documentation

Internship Project | SAP AMS Knowledge Transfer Portal

1. Project Overview

One-line summary:

> *A web portal that manages the transition of SAP AMS support from one team to another — capturing all knowledge, tasks, training, and escalation contacts in a single structured system.*

Problem Statement

When SAP support moves from one vendor or team to another, several critical problems occur:

- Knowledge gets lost — no one documents what was built or why
- Documentation is missing or scattered across emails and drives
- New team members are untrained and unprepared for production support
- Tasks are forgotten with no tracking or accountability
- Important contacts are unavailable during incidents and escalations

The SAP AMS Transition Management System solves all of these problems in one platform.

2. Teams Involved

Team	Responsibilities	Example
Existing Support Team	Write knowledge articles, share documents, explain business processes, transfer knowledge to new team	Article: How Month-End Closing Works — Author: John
Transition Team	Track transition activities, assign tasks, monitor project status, ensure milestones are met	Tasks: Conduct KT Session, Complete UAT, Prepare Go-Live Checklist
New Support Team	Complete assigned training, read knowledge articles, learn SAP processes, become production-ready	Employee Kanan completes SAP FI Basics training
Management Team	Handle escalations, make decisions on critical issues, resolve blockers	L3 escalation contact for Finance module issues

3. Complete Project Flow

This is the end-to-end journey of a transition — from old team to production support:

Existing Team
↓
Creates Knowledge Articles
↓
Transition Team Creates Tasks
↓
New Team Completes Training
↓
Support Team Becomes Ready
↓
Production Support Starts
↓
Issues Occur
↓
Key Contacts & Escalation Used

4. The 4 Functional Modules

The system is built around 4 core modules (Feature Requirements). Each solves a specific part of the transition problem.

FR-2 — Knowledge Base Module

Think of this as the company's Wikipedia for the project. It stores all solutions, documents, procedures, and business knowledge so nothing is ever lost.

Field	Example Value
Title	Invoice Posting Error
SAP Module	Finance (FI)
Content	Step-by-step guide to solve invoice issues in SAP
Author	John (Existing Support Team)
Category	Error Solution

Article Flow:

Old Team Creates Article
↓
Saved to Database
↓
New Team Searches
↓
Reads & Learns the Process

Types of articles:

- Month-end closing process guides
- SAP error solutions and workarounds
- Configuration procedures and setup notes
- Knowledge Transfer (KT) session documentation
- Business process explanations

FR-3 — Transition Checklist Module

This module tracks every activity that must be completed during the transition. Nothing gets forgotten.

Field	Example Value
Task Name	Conduct Finance KT Session
Assigned To	Pragati
Status	In Progress
Due	End of Phase 2
Dependency	Must complete after FI articles are uploaded

Task Flow:

Manager Creates Task
↓
Assigns to User
↓
User Completes Work
↓
Status Updated to Done

Example checklist tasks:

- Conduct KT Sessions for all SAP modules
- Transfer all project documents
- Validate user access and permissions
- Complete User Acceptance Testing (UAT)
- Prepare and sign off on Go-Live Checklist

FR-4 — Onboarding Module

Prepares the new support team to be production-ready. Tracks training completion and certifications for each member.

Field	Example Value
Employee	Kanan

Training Module	SAP FI Basics
Status	Completed
Articles to Read	Invoice Posting, Month-End Close
Readiness	Production Ready

Onboarding Flow:

New Employee Joins
↓
Training Assigned
↓
Articles Studied
↓
Training Completed
↓
Marked as Production Ready

FR-5 — Key Contact Registry

Stores all team leads, managers, and escalation contacts. When an issue occurs in production, the new team knows exactly who to call at each level.

SAP Module	Escalation Level	Contact
Finance (FI)	L1 — First Response	John
Finance (FI)	L2 — If no L1 response	Sarah
Finance (FI)	L3 — Critical Issues	David
MM / Procurement	L1 — First Response	Anita
MM / Procurement	L2 — If no L1 response	Ravi

Escalation Flow:

Issue Occurs in Production
↓
Contact Level 1

↓
No Response → Contact Level 2
↓
Contact Level 3 for Critical Issues

5. Database Design

The database is PostgreSQL. All data is organized into logical groups based on the module it belongs to.

Users

Table	What It Stores
users	Name, email, hashed password, role (Existing Team / Transition Team / New Team / Management)

Knowledge Base Tables

Table	What It Stores
knowledge_articles	Title, SAP module, content, author, category, created date
related_articles	Links between articles that are related to each other

Transition / Checklist Tables

Table	What It Stores
transition_phases	Phase names and descriptions (e.g., Phase 1: Discovery, Phase 2: KT)
transition_tasks	Task name, assigned user, status, due date, phase
task_dependencies	Which tasks must be completed before another task can start

Onboarding Tables

Table	What It Stores
onboarding_members	New employee details, start date, readiness status
training_modules	Training module names and descriptions
training_assignments	Which employee is assigned which training and their completion status

Contact Tables

Table	What It Stores

key_contacts	Contact name, role, SAP module, email, phone
escalation_paths	Ordered escalation levels per SAP module (L1, L2, L3)

6. API Structure

Every API in the project follows the same request-response pattern:

Client Request
↓
Route File (routes/)
↓
Controller (controllers/)
↓
Database Query (PostgreSQL)
↓
Response Sent Back

Example — Login API:

Step	What Happens
POST /api/auth/login	User sends email + password
auth.routes.js	Route receives request, passes to controller
auth.controller.js	Checks password, queries database
PostgreSQL	Returns user record
JWT Token	Generated and sent back to client
Client	Stores token, redirects to dashboard

All routes except /login and /register are protected by the JWT middleware — meaning you must be logged in to access them.

7. Authentication System

Registration Flow

- User fills in name, email, password, and role
- Password is hashed using bcrypt before saving
- User record saved to the users table in PostgreSQL

Login Flow

- User sends email + password
- Backend finds the user by email
- bcrypt compares the sent password with the stored hash
- If match — a JWT token is generated and returned
- Client stores the token (in localStorage or cookie)

JWT Middleware — The Security Guard

Every protected API call goes through this middleware first:

Request Arrives
↓
JWT Middleware Checks Token
↓
Token Valid? → Yes: Continue to Controller
↓
Token Invalid? → Reject with 401 Error

8. Project Architecture

The full stack architecture from what the user sees to what the database stores:

React Frontend (Browser)

↓
Express Routes (API Layer)
↓
Controllers (Business Logic)
↓
Middleware (JWT Auth Check)
↓
PostgreSQL Database

Layer	Technology & Purpose
Frontend	React.js — What Kanan and Pragati will build next. The UI that all 4 user types interact with.
Routes	Express.js route files — one file per module (auth, knowledge, tasks, onboarding, contacts)
Controllers	JavaScript functions — the actual logic: fetch data, update records, return responses
Middleware	JWT token validation — runs before every protected controller
Database	PostgreSQL — all data stored in structured relational tables

9. Current Progress

What Kanan and Pragati have completed so far:

Completed

Category	What Was Done
Database	**PostgreSQL installed, pgAdmin setup, database created, all tables created**
Backend Setup	**Express installed, server.js configured, environment variables set, DB connection working**
Authentication	**Register API, Login API, password hashing with bcrypt, JWT token generation**
Security	**JWT middleware created, protected routes working**
Git & Collaboration	**Repository created, pushed to GitHub, both members connected**

Remaining Work — What Comes Next

Module	What to Build	Priority
FR-2 Knowledge Base	APIs to create, fetch, search knowledge articles. Frontend article list + create form.	High — Build First
FR-3 Transition Tasks	APIs to create tasks, assign users, update status. Frontend task board (Kanban or list view).	High — Build Second
FR-4 Onboarding	APIs for training assignments and status. Frontend onboarding tracker per employee.	Medium
FR-5 Contacts	APIs to add and fetch contacts. Frontend contact directory with escalation levels.	Medium
Frontend — All Dashboards	Login page, role-based dashboards for all 4 user types, navigation, responsive layout.	High — Build in parallel with APIs
Deployment	Host the app so it can be accessed from any browser. Final testing and demo preparation.	Last Step